SHEEPCARE

INTELLIGENT ESTRUS DETECTION SYSTEM

User Manual · Version 1.1.0

WITH THE SUPPORT OF THE EUROPEAN UNION AND THE OPENAGRI PROJECT

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1. Introduction

SheepCare is an intelligent estrus detection system for sheep. It automatically analyses vaginal resistance data files to classify each animal as Estrus or No estrus, simplifying reproductive management on the farm.

The **local (Edge)** version — described in this manual — works completely offline. All data is stored on the device itself. Optionally, it can connect to the **cloud** version to synchronise results with other devices or the central platform.

> **Note:** SheepCare is a decision-support tool. Results should always be interpreted together with the judgement of the responsible technician or veterinarian.

2. System Requirements

Component	Minimum recommended
Operating system	Windows 10 (64-bit) or Windows 11
Processor	Intel Core i5 / AMD Ryzen 5 or higher
RAM	8 GB
Disk space	2 GB free
Web browser	Google Chrome, Edge or Firefox (recent version)

> **No additional requirements:** SheepCare is installed as a native Windows application. It does not require Docker, an Internet connection, or any pre-installed software.

3. Installation

1. Download the installer **SheepCare-Setup-1.0.exe** and double-click it.
2. If Windows displays a security warning, click **"More info"** and then **"Run anyway"**.
3. Follow the setup wizard steps. The installation does not require an Internet connection and completes in **1–2 minutes**.
4. When finished, a **SheepCare** shortcut will appear on the desktop and in the Start menu.

4. First Launch

Double-click the **SheepCare** icon on the desktop. A small progress window will appear showing the startup status:

- **Setting up the database** — occurs only the first time; initialises the local database. This may take up to one minute.
- **Starting database** — starts the database engine in the background.
- **Starting calendar module** — launches the Farm Calendar.
- **Starting SheepCare** — starts the main service.
- **Ready! Opening SheepCare** — the web browser opens automatically.

Subsequent launches take between **15 and 30 seconds**, since the database is already initialised.

> **Application address:** SheepCare always opens at `http://localhost:8000`. You can return to this address from any browser while the application is running.

> **First-time setup only:** the very first time the application starts, a small window will ask you to create the administrator account for **GateKeeper** (the shared login service used by the Calendar module — see 5.7 Calendar and GateKeeper Login). Choose a username and a password of at least 8 characters; you will need them every time you open the Calendar. This step only appears once — on later launches the same credentials are reused automatically.

5. The Application Interface

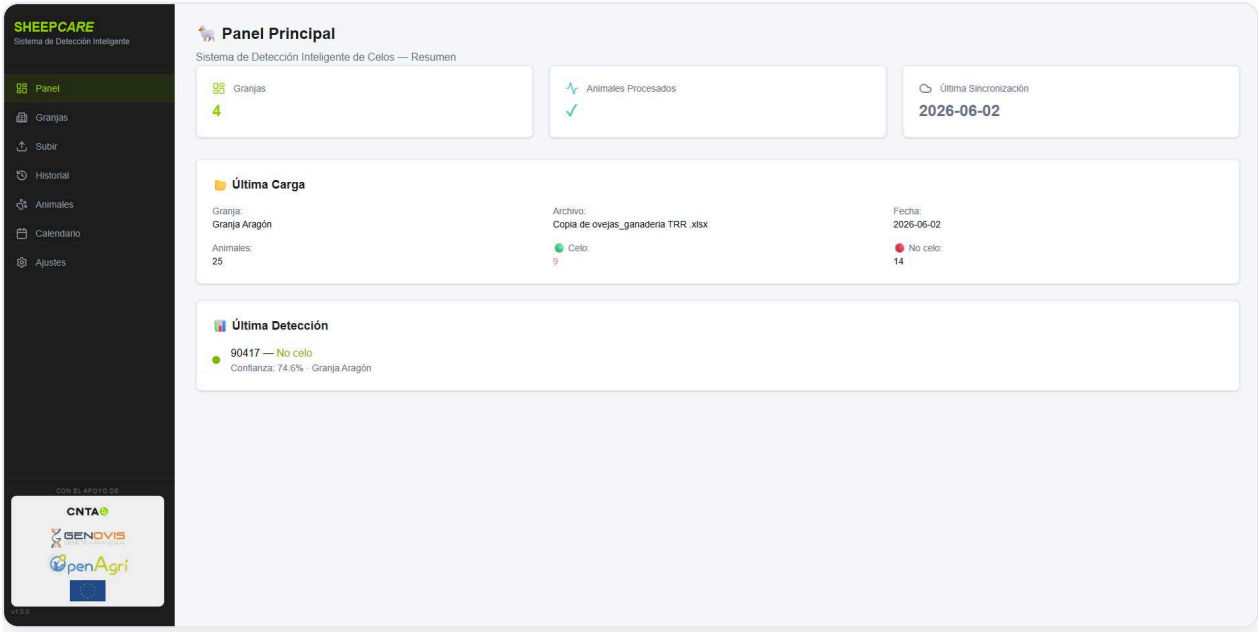
The application is divided into two areas: the **left sidebar** (navigation) and the **content area** on the right.

5.1 Main Dashboard

This is the home screen. It shows a summary of the current farm status:

- **Farms** — number of registered farms.
- **Animals processed** — indicates whether any animals have results.
- **Last synchronisation** — date of the last cloud sync (or "Never" if not configured).

If data has already been loaded, the dashboard also shows a summary of the last upload (farm, file, date, estrus / no-estrus totals) and the most recent detection result.



Main dashboard: 4 registered farms, last upload with 9 estrus detections out of 25 animals. The partner logos are shown at the bottom of the sidebar.

5.2 Farms

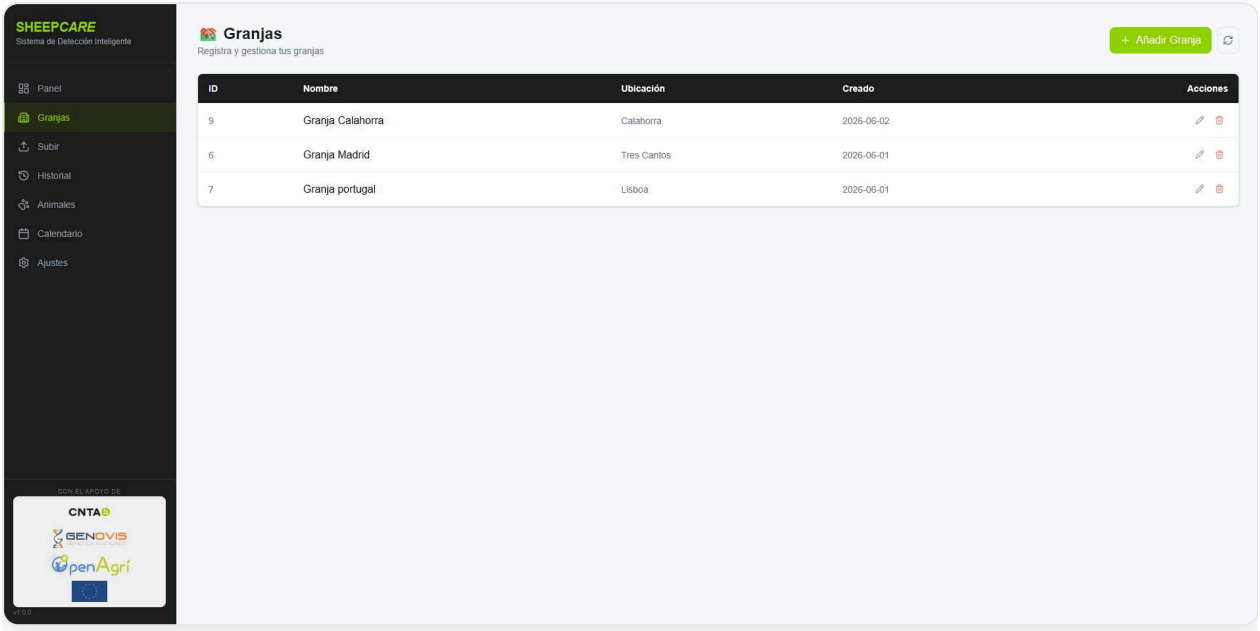
This section manages livestock farms. Before uploading any data file, at least one farm must be registered.

Create a farm

1. Click "➕ **Add Farm**".
2. Enter the **farm name** (required) and, optionally, its **location**.
3. Click **"Create"**.

Edit or delete a farm

In the farms table, use the icons in the **Actions** column: the pencil (✏️) to edit the name or location, and the bin (🗑️) to delete.

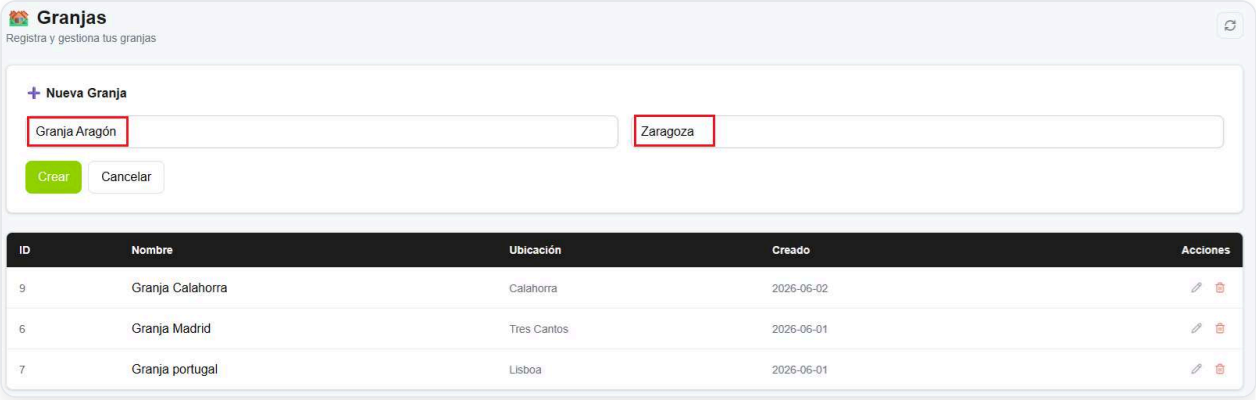


List of registered farms. The ✏️ and 🗑️ icons in the "Actions" column allow editing and deleting each farm.

To register a new farm, click the **"+ Add Farm"** button in the top-right corner of the screen:

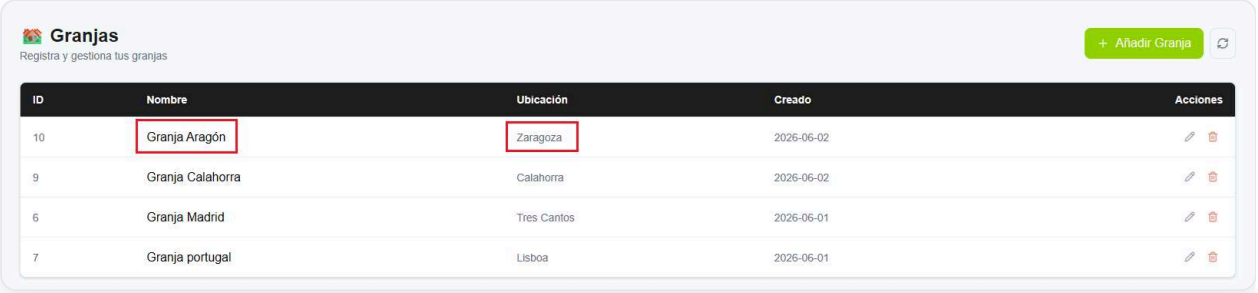
"+ Add Farm" button.

A form will appear over the list. Enter the **farm name** (required) and, optionally, its **location**, then click **"Create"**:



"New Farm" form with the Name and Location fields filled in, ready to save.

Once created, the farm appears immediately at the top of the list, ready for data uploads:



Result after clicking "Create": the new farm (Granja Aragón, Zaragoza) appears instantly at the top of the list.

> **Warning:** Deleting a farm will also delete all its animals, uploads and detection results. This action cannot be undone.

5.3 Upload Data and Detect Estrus

This is SheepCare's main function. It allows you to load a vaginal resistance data file and obtain the estrus detection result for each animal.

File format

The system accepts **Excel (.xlsx, .xls)** and **CSV (.csv)** files with the following structure:

Column A	Column B	Columns C onwards
Animal identifier (tag number)	(reserved)	One column per measurement date, with the resistance value

Upload process

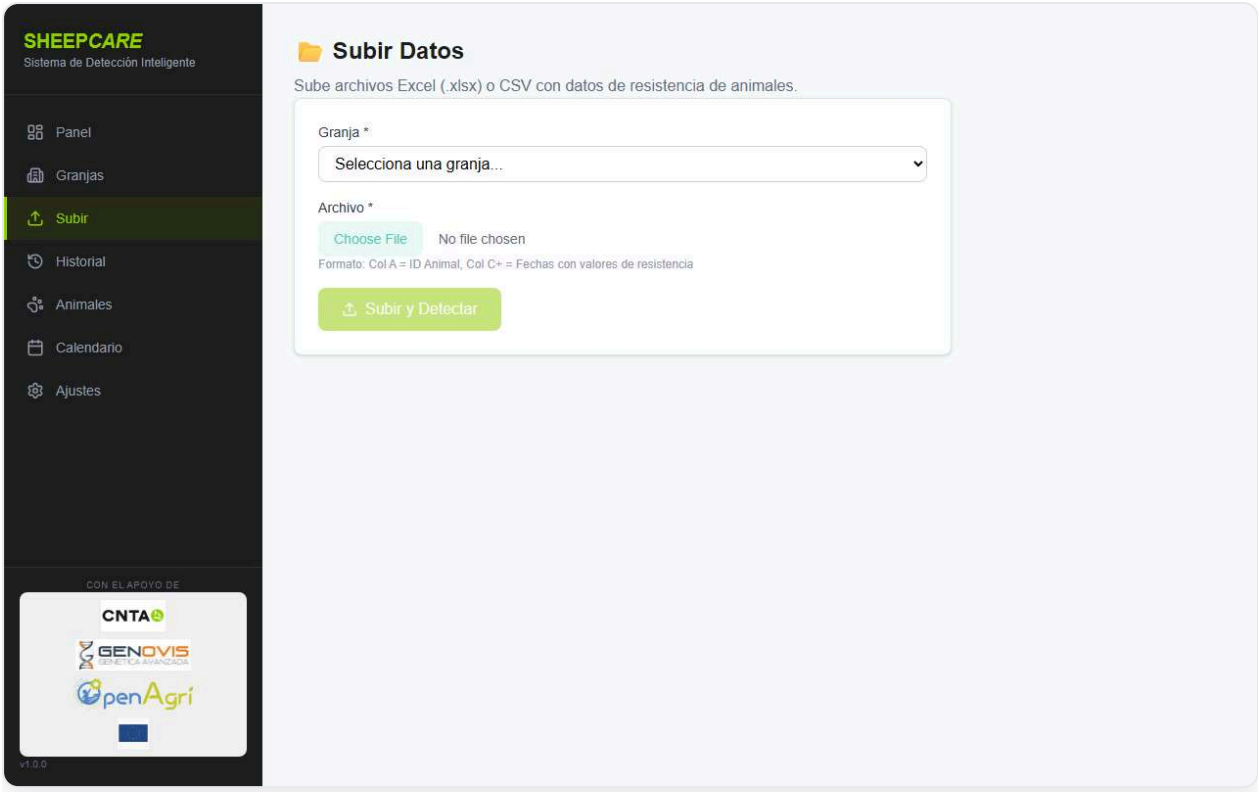
1. In the sidebar, select **"Upload"**.
2. Choose the **farm** to which the animals in the file belong.
3. Click the file field and select the file, or drag it to the indicated area.
4. Click **"Upload and Detect"**. The system will process the file and display the results on screen within a few seconds.

Interpreting results

After processing, a table is shown with one result per animal:

Result	Meaning	Indicator
Estrus	The animal shows signs of being in estrus	Red / coral
No estrus	The animal shows no signs of estrus	Grey

The **Confidence** column shows the model's certainty percentage. Values above 80% indicate a highly reliable result.

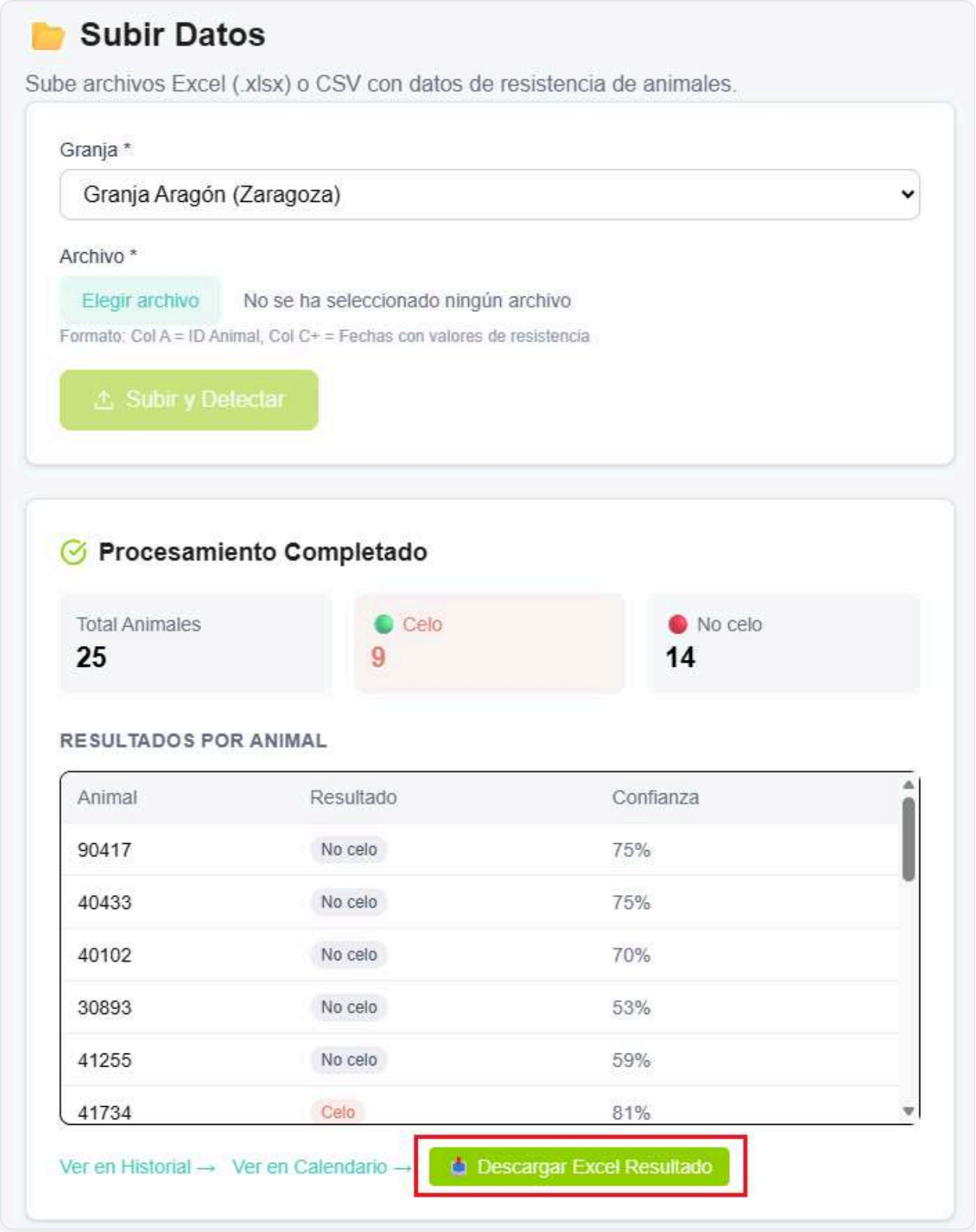


"Upload Data" screen in its initial state: dropdown to select the farm and file selector.

The upload process follows three steps: **(1)** select the farm from the dropdown, **(2)** click "Choose file" and select the Excel file with the resistance data, and **(3)** click "Upload and Detect" to start the analysis:

Form with farm and file already selected. The three steps are highlighted for easy reading.

Within a few seconds the application processes the file and displays the results directly on screen: the total number of animals analysed, how many tested positive for estrus and how many did not, with the confidence percentage for each. From here you can navigate to History, open the Calendar, or download the report:



Analysis results: 9 animals in estrus and 14 without estrus out of a total of 25. The "Download Excel Report" button (highlighted) generates the report file.

Clicking "Download Excel Report" downloads the original file with a new **"Result"** column added at the end. Each cell shows "Estrus" or "No estrus" in the corresponding colour, making it easy to review and integrate into other management systems:

	A	B	C	D	E	F	G	H
1	identificación	Id_secuencial	18/05/2026	19/05/2026	20/05/2026	21/05/2026	22/05/2026	Resultado
2	90417	1	480	330	390	460	390	No celo
3	40433	2	490	310	410	410	430	No celo
4	40102	3	490	270	230	390	360	No celo
5	30893	4	430	330	330	310	370	No celo
6	41255	5	450	310	300	360	390	No celo
7	41734	6	600	430	280	360	450	Celo
8	31662	7	400	300	350	360	310	Celo
9	40356	8	630	250	250	350	340	Celo
10	9577	9	490		380	320	340	
11	40114	10	580	370	480	310	340	Celo
12	40188	11	500	400	280	430	400	Celo
13	42783	12	500	340	320	370	420	Celo
14	30002	13	550	440	380	340	380	Celo
15	31980	14	430	430	370	390	420	No celo
16	42040	15	430	330	380	430	420	No celo
17	40259	16	450	330	250	370	360	No celo
18	01578	17	430	330	300	300	400	Celo
19	41275	18	460		240	360	410	
20	31092	19	440	350	250	330	410	No celo
21	00860	20	410	290	270	330	330	No celo
22	20794	21	320	500	290	460	300	No celo
23	1554	22	590	290	330	330	380	Celo
24	30890	23	330	230	200	390	340	No celo
25	E303	24	440	400	360	340	430	No celo
26	91046	25	400	450	250	330	400	No celo

Downloaded Excel: column H "Result" added automatically with Estrus (in red/coral) and No estrus (in grey) values for all animals in the original file.

Available actions after detection

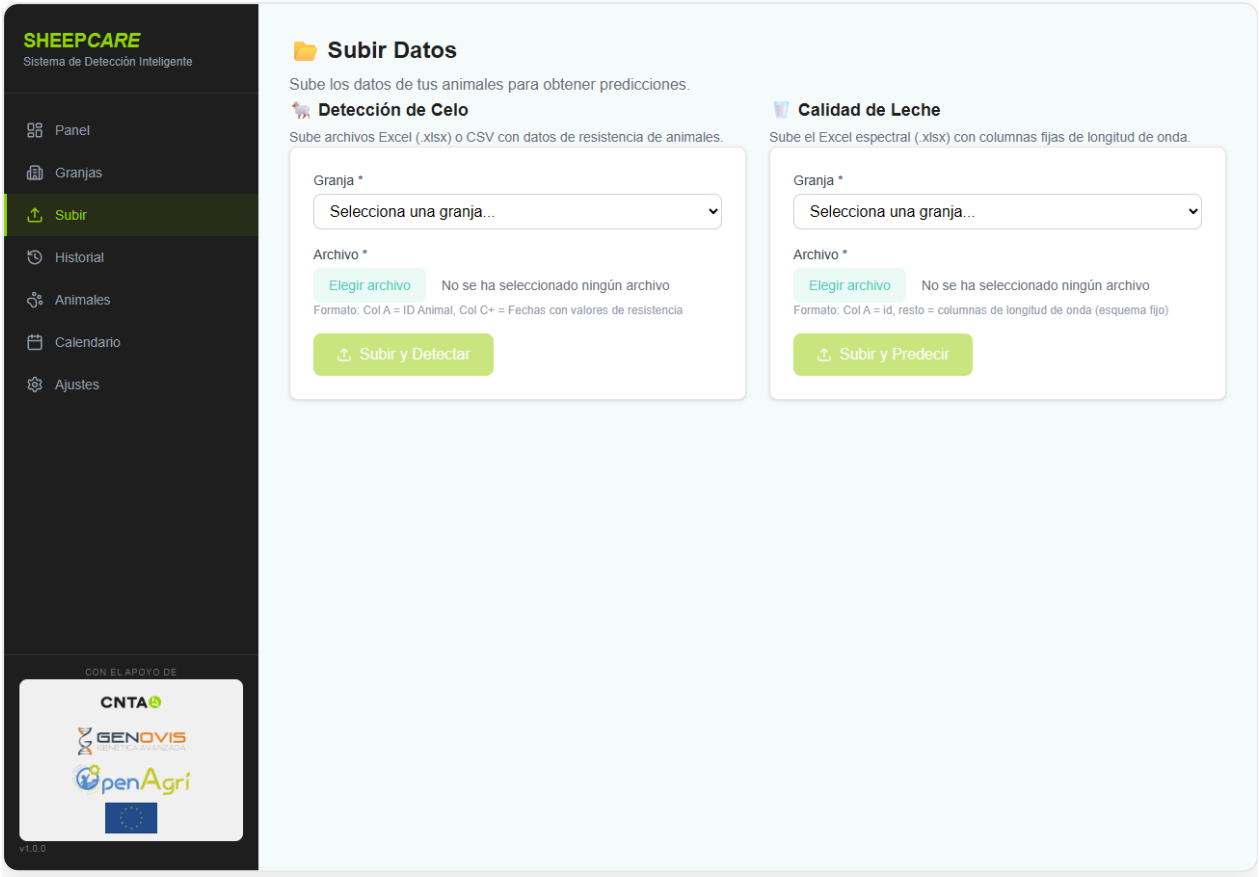
- **View in History** — navigate to the history to consult all results.
- **View in Calendar** — open the management calendar to record events.
- **Download Excel Report** — download an Excel file with all results.

5.4 Upload Milk Quality Data

Next to the estrus upload card, the **Upload** screen has a second card, **Milk Quality**, that classifies milk somatic cell count (SCC) from near-infrared spectral readings, independently of estrus detection.

File format

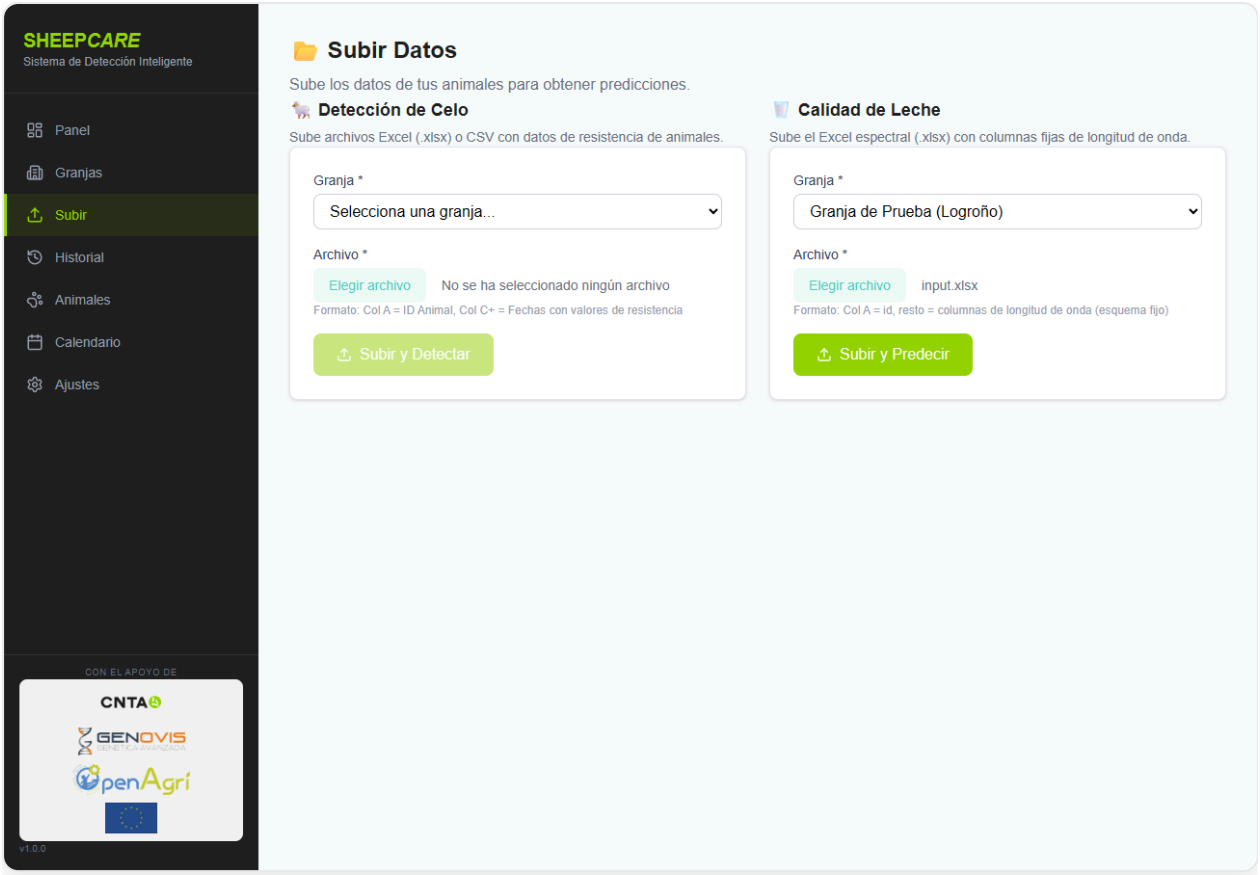
Unlike the estrus file, the spectral file has a **fixed column layout**: the first column must be named `id` and every following column must be one of the exact wavelength values the model was trained on. If the uploaded file does not match this layout exactly, the system rejects it with a clear error instead of producing unreliable predictions.



The "Upload" screen showing both cards side by side: Estrus Detection on the left, Milk Quality on the right.

Upload process

1. In the **Milk Quality** card, choose the **farm** the samples belong to.
2. Click the file field and select the spectral Excel file.
3. Click **"Upload and Predict"**. Processing usually takes a few seconds, even for files with several hundred samples.



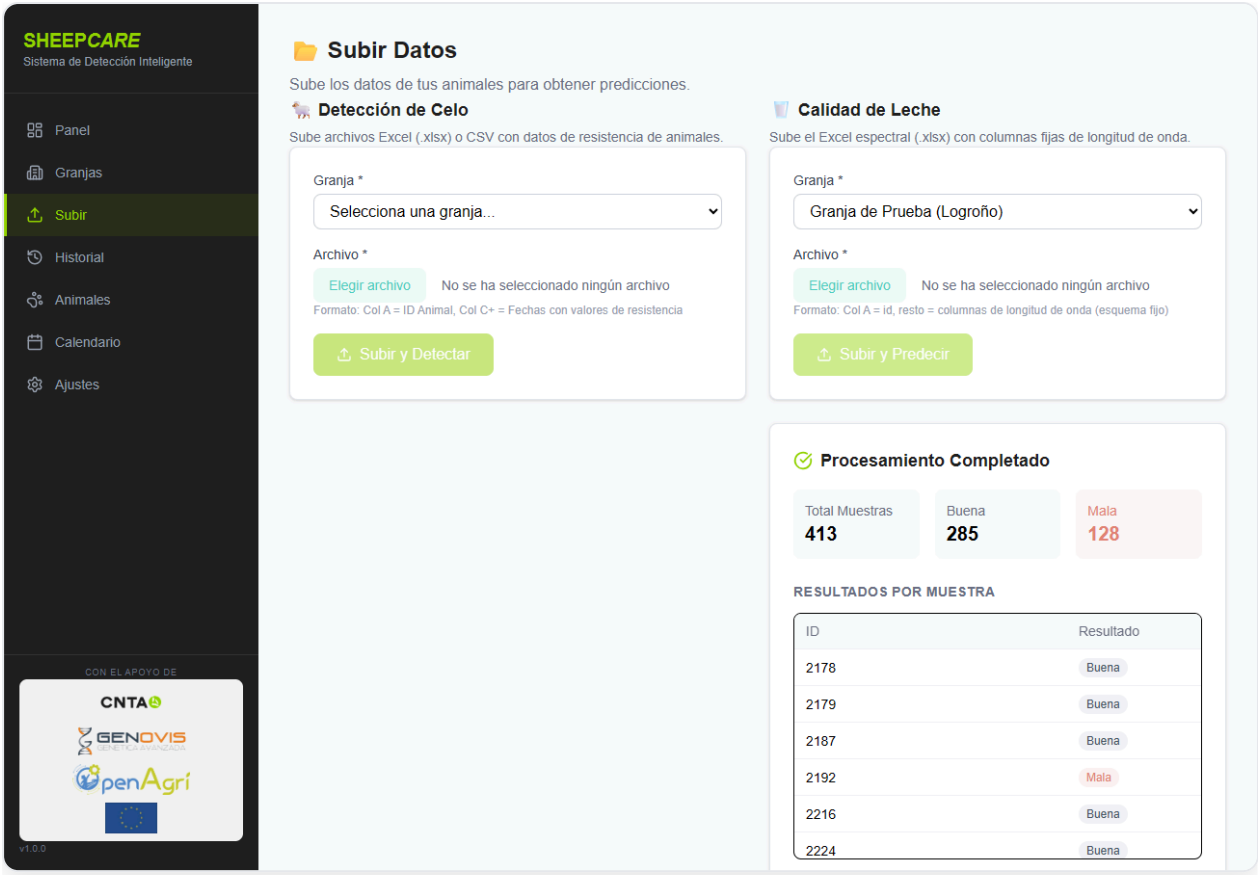
Milk Quality card with a farm and a spectral file already selected, ready to upload.

Interpreting results

Each sample is classified into one of two categories:

Result	Meaning
Buena	Good milk quality — normal somatic cell count.
Mala	Poor milk quality — elevated somatic cell count, worth a closer look.

Unlike estrus detection, this model does not report a confidence percentage for each sample.



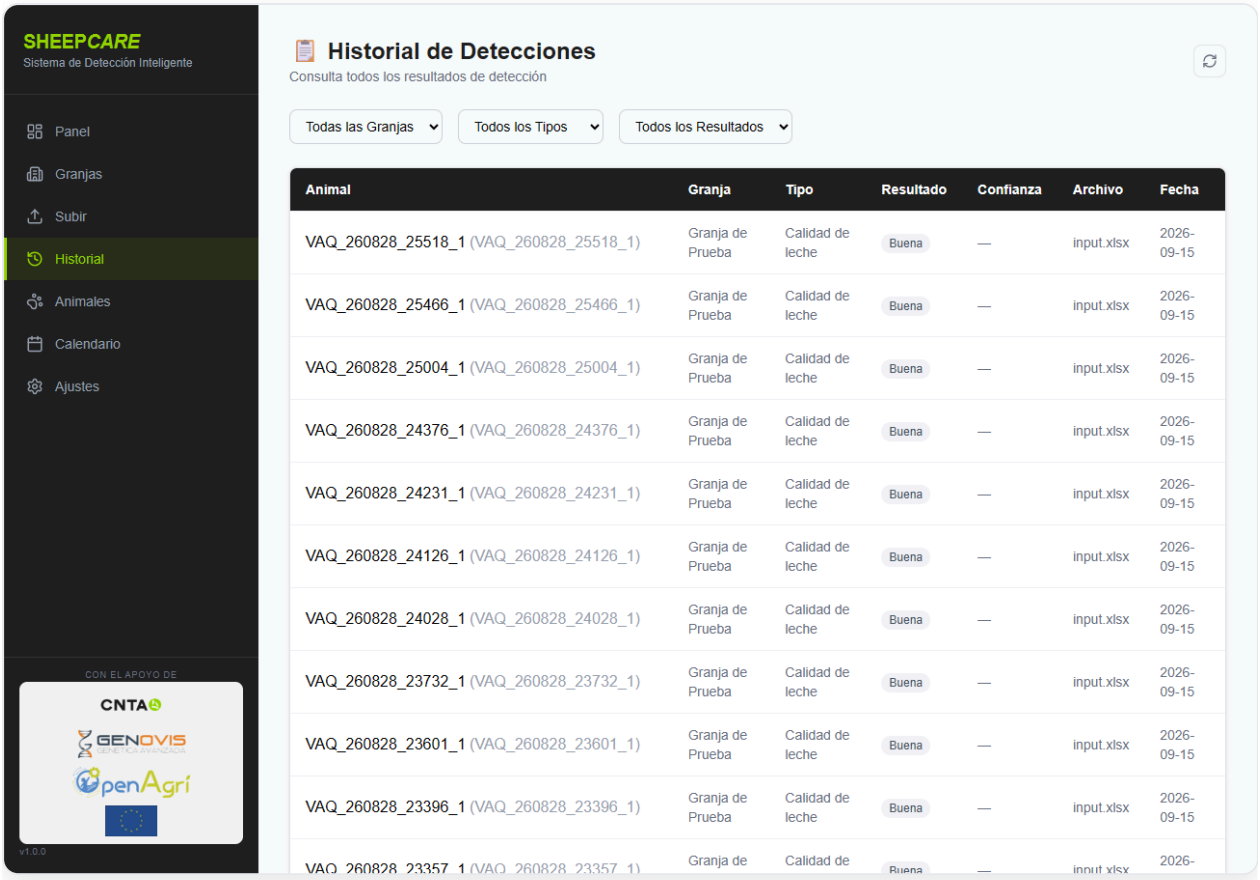
Results for a 413-sample file: 285 classified as "Buena" and 128 as "Mala", with the per-sample breakdown below.

As with estrus results, you can navigate to **History** or download an **Excel report** with a "Resultado" column added to the original file.

5.5 Detection History

Shows all recorded detection results — both **estrus** and **milk quality** — in a single combined list, with the option to filter by **farm**, by **type** (Estrus / Milk quality) and by **result**.

The table includes: animal identifier, farm, type, result, confidence (only available for estrus results), source file name and detection date.

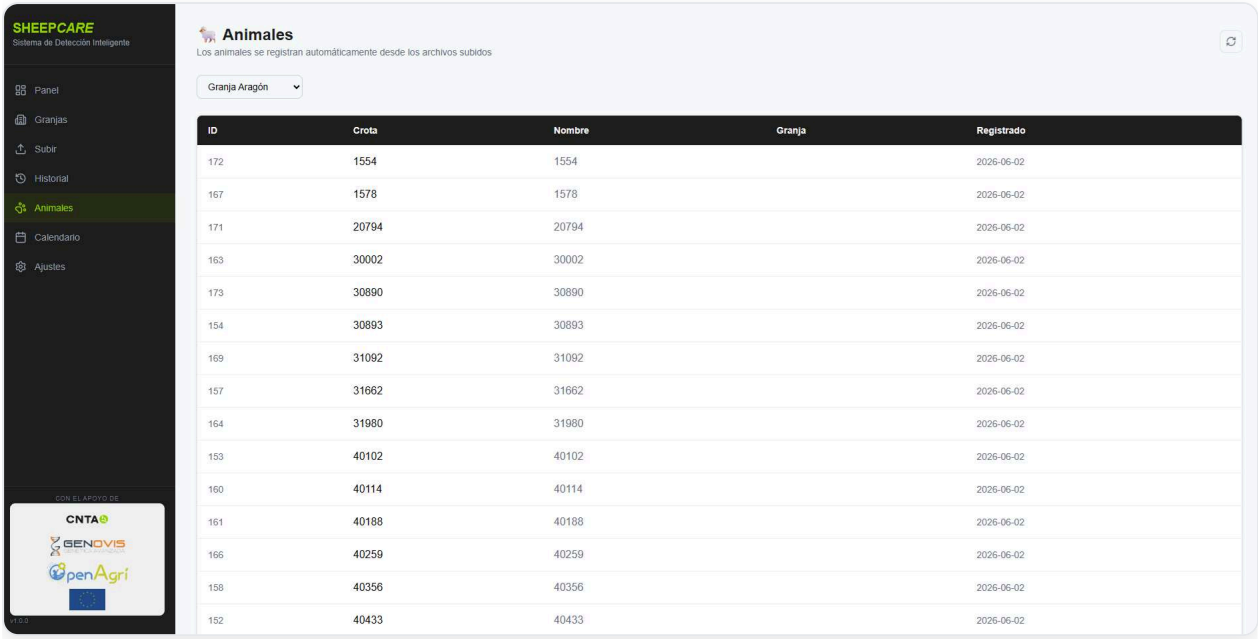


Combined detection history with the "Type" filter. Estrus and milk quality results share the same table, distinguished by the "Type" column.

5.6 Animals

Lists all animals registered in the system. Animals are **not created manually**: they are generated automatically when data files are uploaded, by extracting the identifier (tag number) from each row.

You can filter by farm to view the animals belonging to a specific operation.



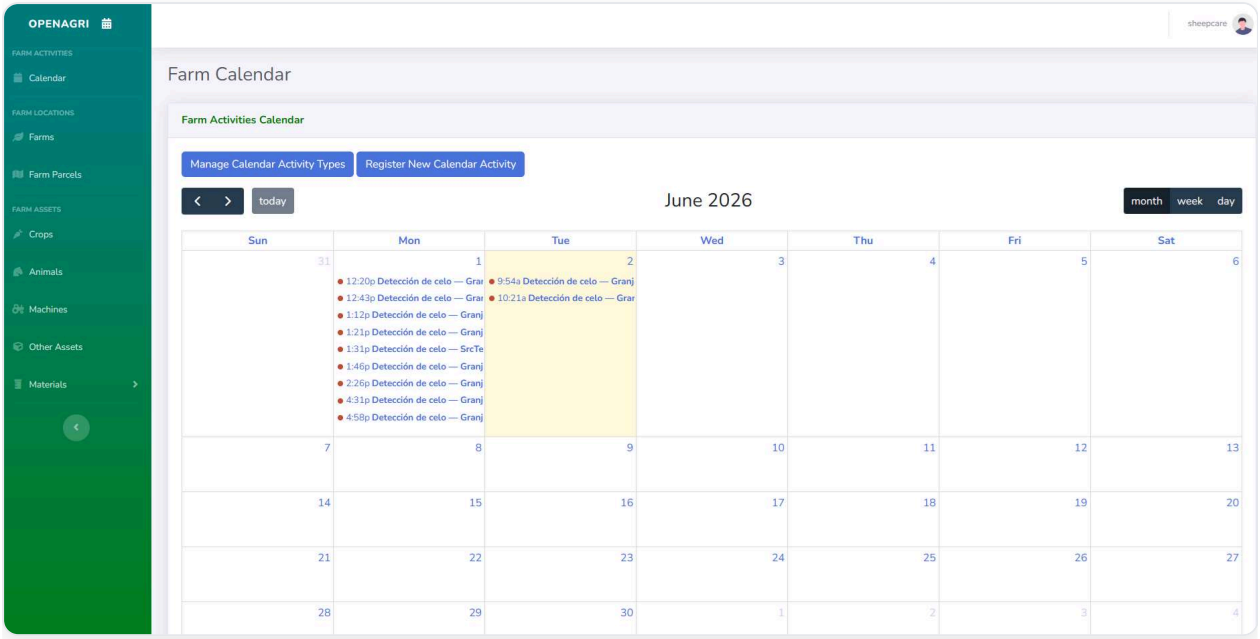
ID	Crota	Nombre	Granja	Registrado
172	1554	1554		2026-06-02
167	1578	1578		2026-06-02
171	20794	20794		2026-06-02
163	30002	30002		2026-06-02
173	30890	30890		2026-06-02
154	30893	30893		2026-06-02
169	31092	31092		2026-06-02
157	31662	31662		2026-06-02
164	31980	31980		2026-06-02
153	40102	40102		2026-06-02
160	40114	40114		2026-06-02
161	40188	40188		2026-06-02
166	40259	40259		2026-06-02
158	40356	40356		2026-06-02
152	40433	40433		2026-06-02

Animal list filtered by farm. Each row shows the animal's tag number and the date it was registered. Animals are created automatically when data files are processed.

5.7 Calendar and GateKeeper Login

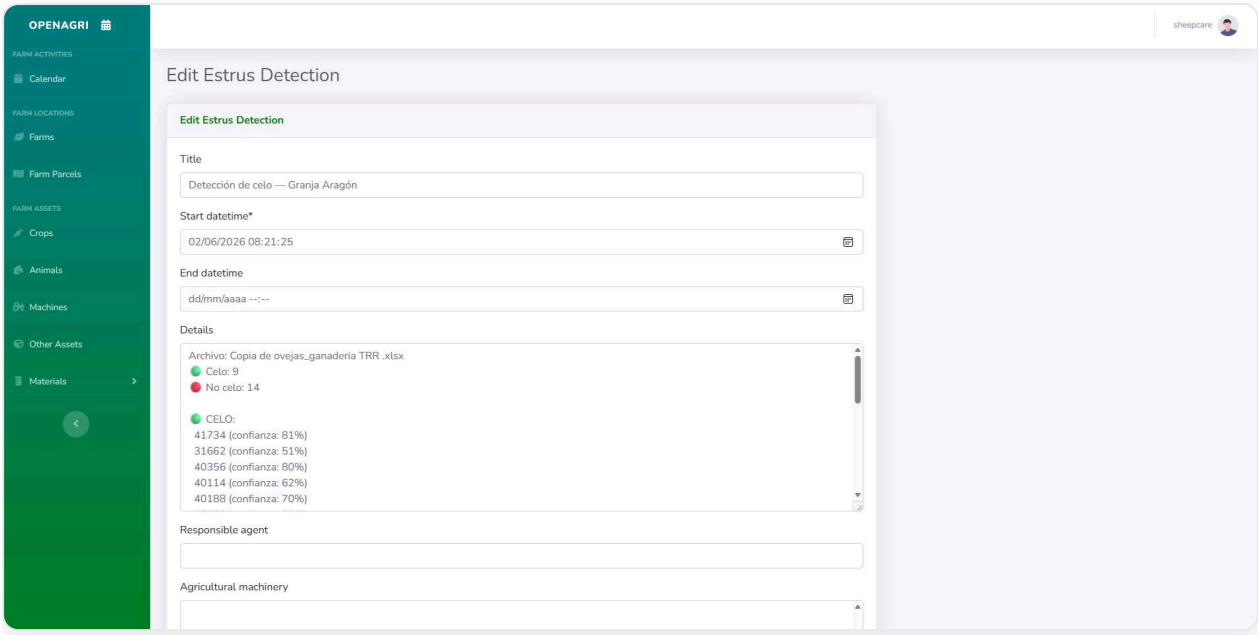
Opens the **OpenAgri Farm Calendar** module in a separate browser tab (`http://localhost:8002`). This module allows you to record and consult farm events (calvings, treatments, matings, estrus detections, etc.) in a visual calendar.

Each time a data file is uploaded in SheepCare, the detection results are automatically recorded in the Farm Calendar as observations.



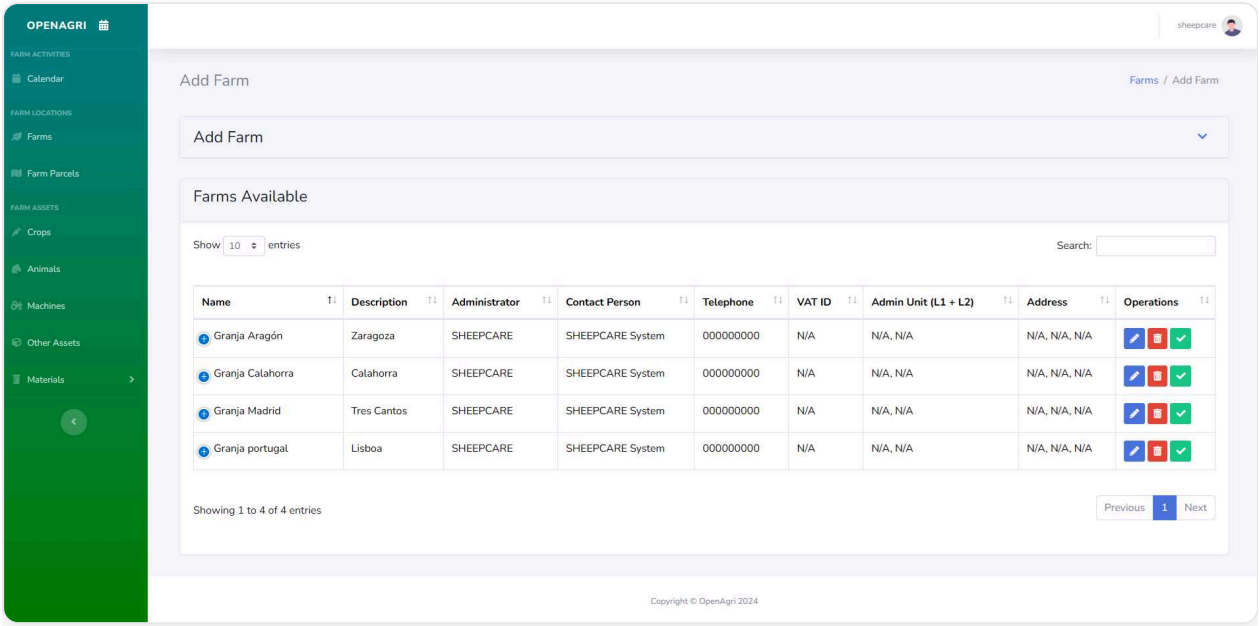
Monthly view of the Farm Calendar. Each data upload processed in SheepCare automatically creates an "Estrus Detection" event on the corresponding day, linked to the source farm.

Clicking on any calendar event opens a detail window with the full information for that detection: associated farm, date and time, and in the **"Details"** field the breakdown of results — number of animals in estrus and not in estrus, and the list of positive tag numbers with their individual confidence percentage:



Event detail window: title with farm, start date, and Details field with the detection summary (9 estrus, 14 no estrus) and the positive tag numbers with their confidence.

From the Farm Calendar sidebar, clicking **"Farms"** gives access to the list of registered operations. By default, this list reflects the same farms created in SheepCare, since synchronisation between the two modules is automatic:

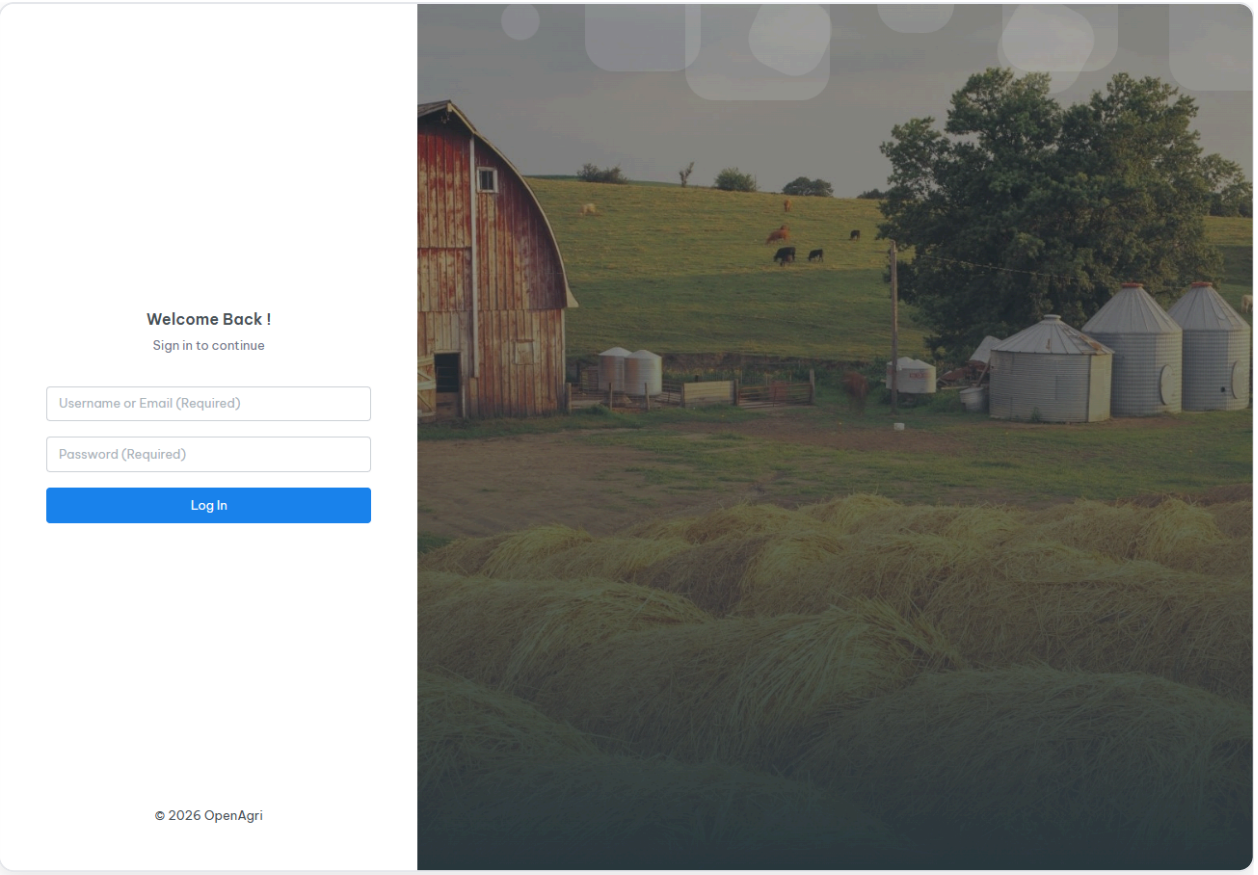


"Farms" section of the Farm Calendar: SheepCare farms (Granja Aragón, Calahorra, Madrid, Portugal) are automatically registered in the calendar module.

> The Farm Calendar starts together with the main application and is accessible while SheepCare is running.

Logging in with GateKeeper

The Calendar no longer uses a fixed username and password. Instead, it delegates login to **GateKeeper**, a shared authentication service used across the OpenAgri ecosystem. The first time you open the Calendar in a browser session, you are redirected to the GateKeeper sign-in page:



GateKeeper sign-in page. Enter the administrator account created on first launch (see 4. First Launch) and click "Log In".

After a successful login, GateKeeper redirects you back to the Calendar, already authenticated — there is no separate Calendar account to remember. The session stays active while the browser tab remains open; if you are asked to log in again later, use the same GateKeeper credentials from the first launch.

> **Forgot your GateKeeper password?** It is stored only on this device (not recoverable remotely). Ask your system administrator to check `%APPDATA%\SheepCare\gatekeeper_admin.key` on this computer, or delete that file to be prompted to create a new administrator account the next time SheepCare starts.

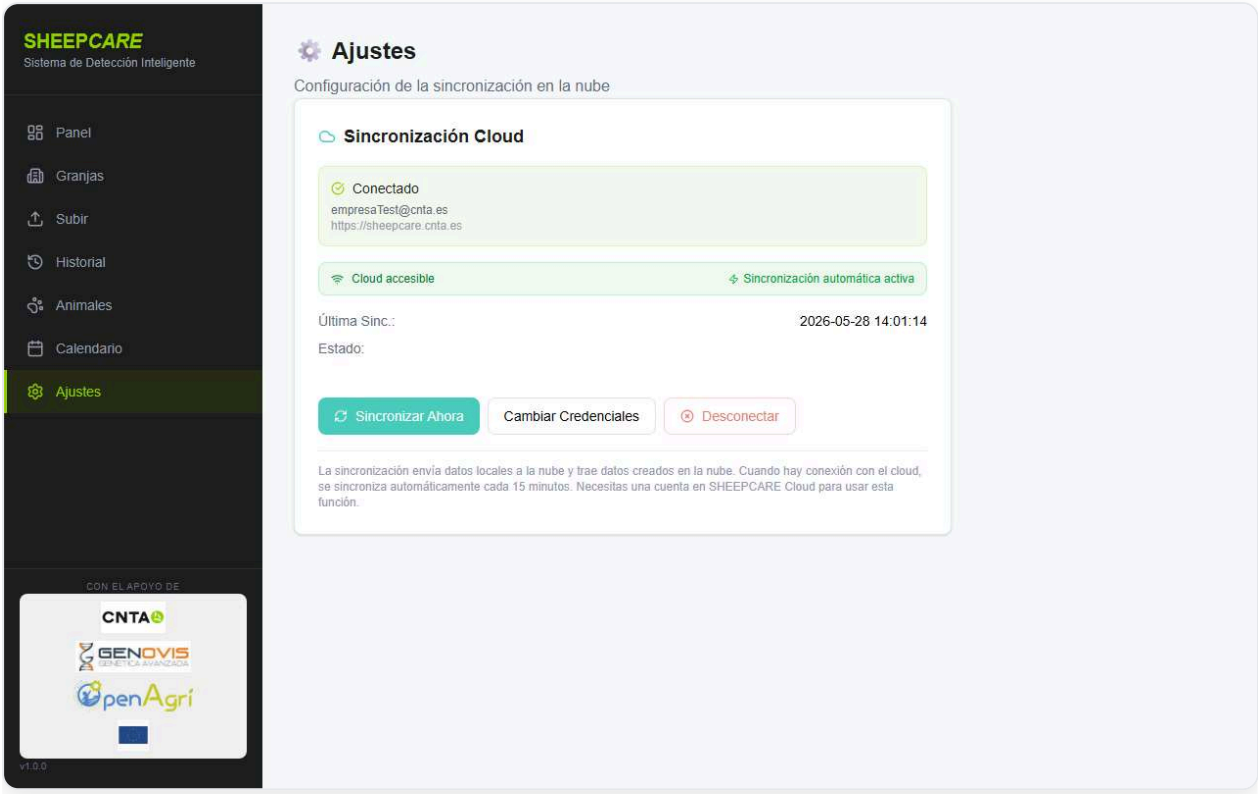
5.8 Settings

Allows you to configure the connection with the SheepCare cloud version to synchronise data between the local device and the central platform.

Connectivity indicator

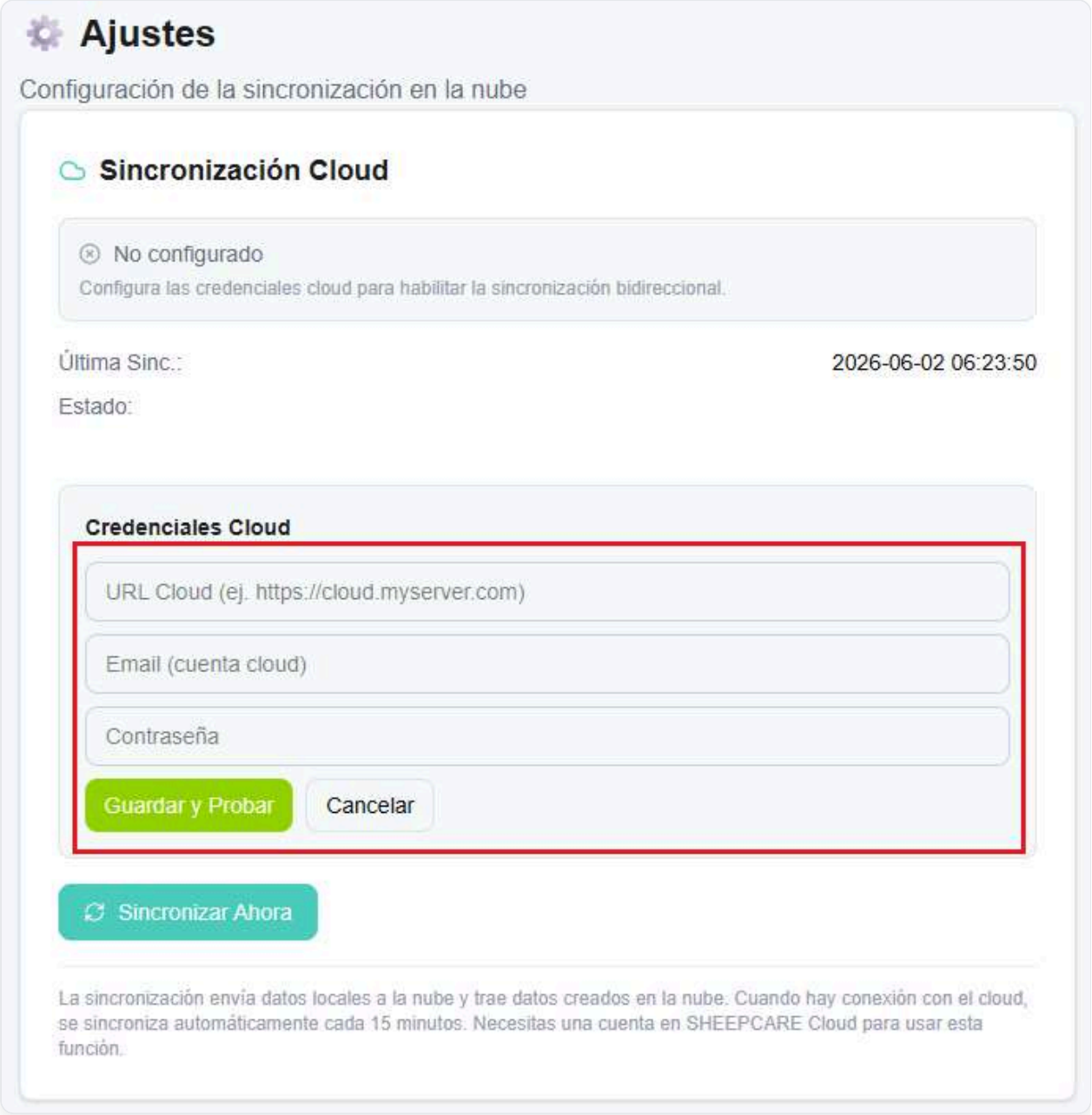
Once the cloud connection is configured, the Settings screen shows in real time whether the cloud server is reachable:

- **Cloud accessible / Automatic sync active** — the connection works and data syncs automatically every 15 minutes.
- **Cloud not accessible — manual sync only** — there is no connection at this moment; data will sync when connectivity is restored or when the manual button is pressed.



Settings screen with active cloud sync: connectivity indicators, date of last synchronisation and available actions.

To enable synchronisation, click **"Configure Cloud"** and enter the credentials for your SheepCare Cloud account. The server URL is `https://sheepcare-pre.cnta.es/`. Also enter the email address and password registered on the cloud platform and click **"Save and Test"**:



Credentials form: cloud server URL, email and password. The "Save and Test" button verifies authentication and launches an immediate sync if the credentials are correct.

If the credentials are valid, the Settings screen will show the **"Connected"** status along with the cloud-accessible and automatic-sync-active indicators. From this point on, data will sync periodically without any further action required:

"Connected" status: account verified, cloud accessible and automatic sync active. The date and time of the last synchronisation performed is shown.

6. Typical Workflow

The typical use of SheepCare follows these steps:

1. **Register the farm** — the first time, create the farm in the *Farms* section.
2. **Prepare the data file** — export from the measurement device an Excel or CSV file with the vaginal resistance values.
3. **Upload and analyse** — in the *Upload* section, select the farm and file, and click "Upload and Detect".
4. **Review the results** — check which animals tested positive for estrus and with what confidence level.
5. **Record in the calendar** — note the corresponding matings or treatments in the calendar module.
6. **Download the report** — export the results Excel file to archive or share with the responsible technician.

7. Cloud Synchronisation

Cloud synchronisation allows data recorded on this device to be available on the central SheepCare Cloud platform as well, and vice versa. It is optional and does not affect the local operation of the application.

Configure the connection

1. Go to **Settings** in the sidebar.
2. Click **"Configure Cloud"** and enter the cloud server URL, your email address and your SheepCare Cloud account password.
3. Click **"Save and Test"**. The application will verify the credentials and, if correct, will launch an **immediate synchronisation** in the background.

Synchronisation behaviour

- **When saving credentials** — an immediate sync is triggered as soon as the credentials are successfully verified.
- **On application start** — if credentials are configured and the cloud is reachable, the app syncs automatically 30 seconds after startup.
- **Automatically every 15 minutes** — while the application is running and there is a cloud connection, synchronisation repeats periodically.
- **Manual** — the **"Sync Now"** button allows you to force an immediate sync at any time.

Synchronisation is bidirectional: data created locally is sent to the cloud, and data created in the cloud is downloaded to the local device.

> If the device has no Internet connection or the cloud server is unavailable, the local application continues to operate normally. Data will sync automatically as soon as connectivity is restored.

8. Frequently Asked Questions

I uploaded a file but no results appear, or there is an error

Check the file format: column A with tag numbers, columns C onwards with numeric resistance values, exactly **5 readings per animal**, and extension .xlsx, .xls or .csv.

The milk quality upload is rejected with a column error

The spectral file requires an **exact** column layout (an `id` column followed by the specific wavelength columns the model was trained on). Files exported from a different device or with edited/reordered columns will be rejected. Use the export as generated by the spectrometer, without modifying the header row.

The Calendar keeps asking me to log in / redirects in a loop

Make sure you are entering the **GateKeeper** administrator credentials created on first launch, not the old Calendar-only credentials from earlier versions. If GateKeeper itself failed to start, the Calendar falls back to its own local login screen instead of GateKeeper's — in that case, restart SheepCare and try again.

The Farm Calendar does not load or shows an error

The Farm Calendar (port 8002) may take a few extra seconds to become available after startup compared to the main application. Wait 20–30 seconds and reload the page.

Will my data be deleted if I uninstall the application?

Uninstalling SheepCare removes the application files installed in `C:\Program Files (x86)\SheepCare\`, but the **user data** (database, configuration, calendar database data) remains in the `%APPDATA%\SheepCare\` folder. To remove it completely, delete that folder manually after uninstalling.

Can I use SheepCare without an Internet connection?

Yes. The local version works completely offline. Cloud synchronisation is optional; if not configured, the application operates in a fully standalone manner.

How do I stop the application?

Closing the browser does not stop the SheepCare services: the backend and the Farm Calendar continue running in the background. To stop the application completely, end the **sheepcare-backend.exe** and **sheepcare-calendar.exe** processes from the Windows Task Manager, or restart the computer.

The background services consume very few resources and do not affect the normal performance of the device. In most cases it is not necessary to stop them manually.